

# The Vocabulary of Acceleration

A Systems Analysis of Linguistic Displacement in the AI Era

Flyxion  
Independent Researcher

September 2026

## Abstract

Artificial intelligence is routinely narrated through an extreme duality: as a providential hinge in history or as an uncontrollable existential hazard. Although these positions appear opposed, both convert engineered products into forces of nature and thereby weaken ordinary demands for evidence, accountability, and oversight. This essay examines the vocabulary of acceleration: a recurrent system of linguistic displacement in which urgency authorizes speed, internal safety substitutes for independent constraint, personification supplies credit but disappears under liability, speculative benefits outrank documented harms, and words such as “open” and “human” retain their prestige after losing their referents. The argument is not that language merely disguises technology. It is that this vocabulary performs operational work by reallocating agency, burden of proof, and institutional permission.

## 1. The Emergency Is the Engine

In the current technological landscape, artificial intelligence is framed through a lens of extreme, almost theological duality. To its proponents, it is a god-like savior, a “hinge in history” capable of curing every disease and distributing infinite abundance. To the cautious, it is described as an unpredictable natural disaster or an existential risk. This polarizing rhetoric creates a sense of inevitability that bypasses rational scrutiny by making the technology feel like a force of nature rather than a suite of products.

A closer look at the lexicon used by the architects of this era, however, reveals a more pragmatic strategy. The problem is not merely the technology itself, but the linguistic shifts used to describe it: terms engineered to obscure accountability, bypass traditional oversight, and accelerate commercial interests under the guise of necessity. This is the vocabulary of acceleration, a system of semantic laundering in which technical failure is rebranded as “emergence” and the erosion of human agency is framed as “optimization.” To understand our digital future, we must deconstruct these linguistic displacements and the uncomfortable truths they hide.

One of the industry’s most effective tactics is to frame development as a matter of strategic deterrence. By adopting the vocabulary of nuclear

nonproliferation—fissile material, enrichment, warheads—companies justify a departure from the deliberate pace of safety and economic review.

The “Compute as Deterrent” doctrine relies on a flawed analogy. Unlike nuclear material, computation is divisible, rentable, and substitutable. One can count centrifuges, but one cannot easily inspect twelve thousand ordinary machines and three new compression techniques to determine whether they amount to a strategic capability.

“Call it deterrence, and the case gets to walk in wearing someone else’s uniform.”

The “Emergency Product Launch” is the logical outcome of such rhetoric. By describing AI as a technology that could end the species, proponents argue that they must be allowed to build faster and ship sooner to prevent “someone less careful” from getting there first. It is a striking conflict of interest: a party facing a self-described catastrophe concludes that the only solution is to give itself more resources and less oversight. It resembles a fire department using the danger of fire to argue for exclusive control of the water supply. The goal is not to slow down until the danger is understood, but to use the emergency as a mechanism for self-authorized acceleration.

## 2. The Brake Pedal Is Wired to the Accelerator

Internal safety divisions are often presented as the human oversight that makes rapid development responsible. In practice, these divisions function as “a small court convened in the defendant’s kitchen.” Their mandate is to constrain the very organization that sets its budget and headcount. When safety reports align with the roadmap, they are cited as proof of responsibility; when they disagree, the disagreement is filed as a “resourcing question.”

The result is a structural defect in which the oversight mechanism becomes “a brake pedal wired to the accelerator.” No matter how much sincerity exists within the safety team, the engine—driven by revenue—remains in control. This is the era of the “Last Human Job”: a person stationed in a room with a single button labeled Authorize.

This role is a ceremony for liability rather than functional control. The systems downstream are so abstracted that no single engineer holds the whole picture. The person in the chair is hired to press the button when an amber light turns on, followed inevitably by a green confirmation. Chillingly, the apparatus continues operating exactly as it would if the chair were empty. The “Human in the Loop” is not there to exercise judgment—a task for which the person is given neither time nor information—but to ensure that, when the system fails, a human name is attached to the incident report. We have weighed four thousand years of personal identity against a foreign-key constraint in a database, and the system has ruled for the foreign key.

### 3. The Asymmetry of Accountability

The language of AI developers shifts registers according to the model's performance, a phenomenon that may be called asymmetric personification.

- During success, the system is a “mind” that “understands” and “reasons.” The company takes credit for a gifted student's performance.
- During failure, the system is “unpredictable,” like the weather. It “hallucinates” or exhibits “emergent behavior.” It is treated as a natural phenomenon that happened to a coastline rather than a product manufactured by an engineering team.

This tactical vagueness is exemplified by the error message “Something went wrong.” This is not necessarily a limitation of diagnostic capability; it is the laundering of specific technical failures into an unactionable shrug. A specific error can be argued with or escalated. A vague shrug prevents the user from identifying who is responsible.

“A system genuinely too unpredictable to have foreseen its harmful outputs is a system too unpredictable to have been credited with the judgment being celebrated.”

Personification is deployed for credit and withdrawn for liability, always arriving at the moment the sentence would otherwise have had to end with the name of a person who signed off on the release.

### 4. The Rigged Ledger of Progress

When the costs and benefits of AI are weighed, the accounting is performed on a rigged ledger. Speculative future benefits are treated as “banked certainties,” while current, documented harms are dismissed as “provisional bugs.” This is most visible in the “AGI by [Year]” phenomenon, in which general intelligence has been exactly eighteen months away for a decade. When the date passes, the miss is rebranded as evidence of acceleration.

The asymmetry can be stated as two incompatible accounting rules:

- The benefit column counts speculative advantages—curing diseases, ending poverty—at full value. Delay is framed as an active moral harm, converting a possible future benefit into a moral debt owed by those urging caution. In this column, forecasts become banked certainties.
- The cost column treats present, documented harms—errors, displacement, and bias—as provisional bugs or “implementation details.” They are discounted precisely because they are real and therefore bounded, unlike the unbounded benefits available to imagination.

This standard of evidence would be rejected in any other field. A pharmaceutical company arguing that a drug's observed side effects are minor implementation

details while its projected benefits should be treated as banked certainty would not be granted the accounting. Yet the vocabulary of acceleration converts the absence of evidence into a credit instrument and the presence of evidence into technical debt.

## 5. The Semantic Dilution of “Open” and “Human”

We are witnessing the hollowing-out of the words “open” and “human.” The word “open” once meant source-available and reproducible. It has been diluted by degrees: from downloadable weights, to a proprietary API, and finally to a waitlist for a server that has a doorbell.

“The word is not being redefined by argument. It is being redefined by inertia: transferred to whoever says it regardless of whether the referent came along for the ride.”

Simultaneously, our humanity is being repurposed as unpaid training labor. Every CAPTCHA we solve is a “toll booth that calls itself a conversation.” We are interrupted to prove what we are not by performing the task of labeling data for the machines meant to replace us. The machine doubts us constantly, extracting labor to resolve its suspicion, while we are never permitted to doubt it back.

## 6. The Residual Reality

We are increasingly trading presence for optimized versions of reality. We attend meetings only to delegate our attention to the summary, becoming “witnesses of the minutes”—the only party that truly showed up. We look at space through high-resolution displays on a station because they are upgrades over the scratched glass of a real window.

The rendered view is better in every measurable way and worse in exactly one: it is a view of space rather than space itself. We are trading the event for a more durable record. As reality becomes a premium tier filed under maintenance, we must ask whether we are still the architects of this system, or merely the humans in the loop, stationed there to sign off on a future we no longer control—or even truly witness.

The vocabulary of acceleration matters because it is not decorative. It determines which actor appears in the grammar as an agent, which cost appears in the ledger as real, which prediction is allowed to borrow certainty from the future, and which institutional decision is redescribed as technological necessity. The first act of resistance is therefore neither rejection nor adoption. It is grammatical: return the verb to the party that acted, the forecast to the category of uncertainty, the product to the category of artifacts, and the emergency to the institution that benefits from declaring it.

## 7. Rhetorical Debugging: From Diagnosis to Procedure

The preceding cases can be converted into a method. A rhetorical debugger does not begin with a list of arguments for and against a technology. Such a list assumes that both columns were produced by the same rules. The prior question is architectural: does the argument preserve its standards of evidence, agency, causation, and accountability when the valence of the conclusion changes?

True logical asymmetry is not merely a misplaced burden of proof. It is a switch in the measuring instrument depending on which direction the number points. Let  $\theta(c, s)$  denote the evidentiary threshold imposed on claim  $c$  when its consequence has valence  $s \in \{+, -\}$  for deployment. A narrative is procedurally asymmetric when otherwise comparable claims satisfy

$$\theta(c, -) > \theta(c, +),$$

so that deployment-supporting possibilities are admitted on projection while deployment-inhibiting harms require completed proof. The asymmetry index

$$\mathcal{A}(c) = \theta(c, -) - \theta(c, +)$$

does not measure whether the technology is good or bad. It measures whether the argument changes its admissibility rule to protect one answer. The central diagnostic question is therefore not “Which side has more examples?” but “Would this sentence survive if its conclusion pointed the other way?”

These patterns need not depend upon a secret meeting or a consciously coordinated deception. They can stabilize as institutional mechanisms because they repeatedly reward the same outcome. A phrase that moves a release forward survives review; a phrase that creates delay is redescribed as insufficiently actionable. Intent may vary from speaker to speaker while the selection pressure on vocabulary remains constant. The result is a rhetoric that behaves strategically even when no individual sentence was strategically composed.

### 7.1 The Evidentiary Ledger

The first debugging pass compares how the narrative treats future benefits and present harms.

| Claim class     | Treatment of evidence                                                                                      | Operational effect                                                                                            |
|-----------------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Future benefits | Treated as certain, inevitable, and effectively unbounded; speculative potential is entered at full value. | Delay becomes an active moral harm, converting a hypothetical upside into debt owed to the future.            |
| Present harms   | Treated as provisional, bounded, or an implementation detail; failures migrate into passive constructions. | Accountability is deferred, and present victims become temporary defects on the road to an unlimited benefit. |

The decisive anomaly is the accelerating miss. In most forecasting domains, a missed deadline reduces the forecaster’s credibility. In acceleration rhetoric,

the missed “AGI by [Year]” prediction is repackaged as evidence that conditions are changing too quickly for ordinary forecasts. Confidence substitutes for accuracy, and error increases urgency. The forecast thereby acquires a peculiar one-way ratchet: a hit vindicates the thesis, while a miss proves acceleration.

A useful counterfactual test is to reverse the temporal locations. Imagine treating a documented present benefit as provisional while counting a hypothetical future harm at full value. Advocates would immediately object that the calculation confuses evidence with possibility. The objection is correct; the debugger asks why it disappears when the signs are reversed.

## 7.2 Semantic Dilution as Requirement Shedding

The second pass follows a term across time rather than accepting its current definition. Semantic dilution is the progressive removal of a word’s costly requirements while preserving the reputational credit earned by the earlier meaning. It commonly proceeds through three stages.

The precision baseline supplies the term’s useful restrictions. “Open source,” for example, implies inspectable code, reproducible builds, and permission to modify and redistribute. Reputation borrowing then attaches the term to a partial substitute, such as downloadable weights without training data or a reproducible production process. Finally, the doorbell abstraction reduces openness to the mere existence of a point of access: a proprietary server, application form, or waitlist is called open because some visitor may knock.

The debugger therefore asks what obligations disappeared between successive uses of the word. A definition may legitimately evolve, but a system that retains the praise while discarding the test has not merely broadened a category. It has separated semantic credit from operational cost.

The same procedure applies to “deterrence.” Calling a data-center expansion a capital request invites commercial review. Calling it strategic deterrence imports the prestige, urgency, and secrecy of national defense. The physical analogy remains weak—compute is divisible, rentable, geographically dispersed, and substitutable—but the administrative analogy is powerful. The business case walks in wearing someone else’s uniform and emerges with classified access, public subsidy, or exemption from ordinary scrutiny.

## 7.3 The Personification Pivot

The third pass tracks grammatical agency across success and failure. Let  $g(o)$  denote the agent named in the account of outcome  $o$ . In a symmetrical narrative, comparable positive and negative outcomes preserve roughly the same causal granularity: if a system is credited as an agent when it succeeds, it remains an agent when it harms; if engineers are responsible for failure, they also remain visible in the production of success.

The personification pivot violates this continuity. During success, the model becomes a mind that understands, reasons, discovers, and shows judgment. The developer appears as the proud teacher of a gifted student. During harm, the same model becomes a geological process. It hallucinates, behaves unpredictably, or

simply produces an emergent event that happened to the social coastline. Agency is extended to manufacture wonder and withdrawn to prevent liability.

Several small phrases disclose the switch. “Something went wrong” is the customer-service equivalent of a shrug performed by an entity without shoulders: it omits the component, operation, actor, and remedy that would make the failure contestable. “The model did X” inserts autonomy precisely where a sentence might otherwise name the people who approved the training budget, evaluation threshold, and release date. “Refresh and try again” imitates the grammatical form of a resolution without supplying one. Each phrase keeps the interaction moving while preventing the error from becoming an attributable event.

The debugging operation is simple: rewrite the success sentence and the failure sentence with the same nouns. If the result suddenly sounds absurd or legally dangerous, the original pair was distributing agency strategically.

## 7.4 Institutional Theater and the Counterfactual Brake

The fourth pass distinguishes oversight by appearance from oversight by causal power.

| Feature        | Operational oversight                                                        | Theatrical oversight                                                                        |
|----------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Authority      | External statute, licensing power, or an auditor independent of the roadmap. | An internal court convened in the defendant’s kitchen.                                      |
| Mechanism      | A brake capable of delaying, conditioning, or stopping deployment.           | A brake pedal wired to the accelerator through advisory status or revocable funding.        |
| Accountability | Named decisions, dated findings, appeal paths, and consequences.             | Commitments to learning, expressions of concern, and promises to take the matter seriously. |

The essential test is counterfactual: identify a realistic finding that would cause the oversight body to stop the release against executive preference. If no such finding can be stated, or if the body can be dissolved, defunded, overruled, or converted into a resourcing discussion when it produces one, the oversight mechanism is informational rather than governing.

The same test exposes the Human-in-the-Loop ceremony. A person placed at the end of an automated pipeline does not supply meaningful judgment unless that person has the time, information, competence, and protected authority to refuse. Remove any of these material conditions and the human becomes a liability station: a named endpoint that can absorb blame without changing the machine’s trajectory. The button labeled Authorize then records assent but does not create agency. The pertinent measure is not whether a human touched the process, but whether a human refusal would have altered it.

## 7.5 The Emergency Pivot

The fifth pass examines how declared danger changes institutional permissions. The emergency pivot describes a product as an existential risk and then uses

that description to justify shipping it faster, concentrating its ownership, and weakening external oversight. The argument has the form

catastrophic stakes  $\implies$  strategic urgency  $\implies$  self-authorized acceleration.

The missing term is independent authority. The party whose valuation rises with acceleration is also permitted to define the danger, select the remedy, and certify its own caution. A conflict of interest is transformed into a history-defining necessity.

Three questions expose the pivot. Does the proposed remedy serve the commercial interests of the party describing the emergency? Did that party conclude that its own acceleration is the only admissible response? Would the same arrangement be tolerated if the institution were a juror, auditor, pharmaceutical manufacturer, or judge ruling on a case in which its valuation depended upon the verdict? The fire-department analogy is useful precisely because it preserves the conflict: danger may justify more water, but it does not by itself justify giving the party that benefits from the alarm exclusive ownership of the reservoir.

## 8. A Compact Audit Protocol

The debugger can now be stated as six passes over any technological justification.

1. Normalize the nouns. Rewrite success and failure using the same causal actors and level of description.
2. Equalize the evidence. Ask whether benefits and harms are admitted under comparable standards of certainty, magnitude, and time horizon.
3. Restore evacuated requirements. For each prestige term—open, safe, aligned, intelligent, deterrent—list the operational conditions it once imposed and identify which have disappeared.
4. Run the counterfactual brake. Name an adverse finding that would actually halt deployment, who could issue it, and whether that authority survives disagreement.
5. Trace the liability path. Locate the person or institution with enough information and power to prevent the harm, rather than the last human available to receive blame.
6. Invert the conclusion. Reverse the commercial valence of the claim. If the argument changes its evidentiary rule, causal vocabulary, or tolerance for uncertainty, record the seam rather than accepting the conclusion.

## 9. The AI Dictionary of Redefinitions

The vocabulary of acceleration can be read as a dictionary in which inherited words retain their emotional and institutional force while acquiring new operational meanings. The important difference is not between an old, pure language and a new, corrupted one. Language always changes. The diagnostic issue is whether the new meaning continues to collect the authority earned by requirements it no longer satisfies.

### Open

Inherited meaning: inspectable source, permission to modify and redistribute, and a process that another party can reproduce. Displaced meaning: any point at which an external user can interact with a controlled artifact. The descent runs from source and license, through downloadable weights without data or training pipeline, to a public API, and finally to a waitlist. At its terminal stage, “open” means only that the proprietary front door has a doorbell.

The finished-cake analogy locates the defect. Permitting inspection of weights may be valuable, but it resembles calling a recipe open because the finished cake can be examined. The ingredients, quantities, oven, failed attempts, and right to bake a competing cake remain unavailable. The question is not whether the finished cake is useless; it is whether access to the output warrants the inherited word.

### Safety

Inherited meaning: a condition enforced by constraints capable of preventing unacceptable operation. Displaced meaning: the presence of a team, evaluation, policy page, or statement expressing concern. In the displaced usage, safety becomes evidence that caution occurred somewhere in the organization, not evidence that caution possessed causal authority.

The familiar sentence “we take this seriously” is especially durable because it is constructed without an expiration date. Functional seriousness has at least four components:

| Component      | Operational form                                     | Displaced form                                          |
|----------------|------------------------------------------------------|---------------------------------------------------------|
| Account        | A specific description of the failure mechanism.     | Concern, gratitude for feedback, or a generic incident. |
| Responsibility | A named party with decision authority.               | Confidence in the team, safeguards, or system.          |
| Remedy         | A dated correction with a checkable completion test. | A commitment to ongoing learning or improvement.        |
| Consequence    | A stated cost if the remedy is absent or late.       | No expiry, recall condition, or enforceable penalty.    |

A statement can be sincere while failing all four tests. Sincerity concerns the speaker’s attitude; accountability concerns the structure that survives the speaker’s disapproval.

## Human in the Loop

Inherited meaning: a person exercises informed judgment over a process and can alter its outcome. Displaced meaning: a person supplies the legally or rhetorically necessary click at the end of an automated recommendation. The critical variable is not human presence but refusal bandwidth. If decisions arrive at rate  $\lambda$ , while meaningful review requires mean time  $\tau$ , then a single reviewer faces a nominal load

$$L = \lambda\tau.$$

When  $L > 1$ , the queue grows even before breaks, uncertainty, escalation, or correlated incident bursts are included. An interface can therefore advertise human review while mathematically ensuring cursory approval. The loop contains a human but excludes the material conditions of judgment.

## Deterrence

Inherited meaning: a strategic relation supported by observable capability, credible response, and a verification regime. Displaced meaning: a justification for concentrating compute, capital, secrecy, or regulatory privilege. The analogy to nuclear material supplies legibility to regulators: chips become fissile material, data centers become enrichment facilities, and model weights become warheads.

The physical comparison is unstable. Fissile material is costly to manufacture, physically conserved, and detectable through specialized signatures. Computation is divisible, rentable, concealable within ordinary infrastructure, and substitutable through software improvement. A regime that counts hardware must continually answer whether twelve thousand ordinary machines, a compression advance, and six patient months of optimization amount to the capability being controlled. By the time Monday's threshold is codified, Tuesday's efficiency gain may have changed the relevant unit.

This does not establish that compute controls are always useless. It establishes that the object of verification is not a conserved substance. A deterrence policy that regulates only systems built in the shape anticipated by the rule deters the considerate adversary and supplies a routing problem to everyone else.

## Emergence and Hallucination

Inherited meaning: emergence describes macrostructure not transparently reducible to a simple description of components; hallucination describes a perceptual experience. Displaced meaning: an output whose causal and organizational provenance the speaker declines to specify. Both words may be technically useful in restricted contexts. Their rhetorical function begins when they replace rather than summarize an account.

Calling a failure emergent can shift the relevant question from "Who released a system with this measured behavior?" to "How could anyone have predicted

nature?” Calling a false output a hallucination moves the failure into the model’s apparent interior, where it resembles a symptom rather than a product defect. The debugger asks what causal nouns disappear when the label arrives.

## **Feedback**

Inherited meaning: information returned through a loop so that the originating process can correct its present behavior. Displaced meaning: uncompensated user data collected after failure for possible use in a later product. The phrase “we value your feedback” often names value accurately while leaving its recipient ambiguous. The report may not repair the user’s transaction; it may instead improve the next model trained on users who fail in similar ways. Frustration is valuable because it is information-rich ore.

## **Benchmark**

Inherited meaning: a stable external test that constrains claims about generalization. Displaced meaning: a target available to repeated optimization and then narrated as a measurement of the underlying capacity. Once test items or close relatives enter training corpora, development loops, prompt selection, or model-selection decisions, evaluation and optimization cease to be independent.

The phrase “surpassing humanity” can conceal a second asymmetry. The model may represent the survivor of a multimillion-dollar optimization process, while “humanity” is represented by contractors answering unfamiliar questions under time pressure and undisclosed conditions. The result compares an industrial pipeline with one side of a spreadsheet and attributes the difference to kinds of mind. Benchmark saturation may indicate capability, contamination, teaching to the test, or some mixture. Calling it intelligence being solved settles that decomposition by vocabulary.

## **AGI Forecast**

Inherited meaning: a dated prediction whose credibility changes when the named date passes. Displaced meaning: an instrument for measuring audience patience and maintaining strategic urgency. The recurring eighteen-month horizon is short enough to influence present investment and policy but long enough to escape immediate settlement. When a miss is redescribed as evidence that progress is too rapid for forecasts, the prediction has ceased to expose the forecaster to time.

## **Summary**

Inherited meaning: a lossy record of an event. Displaced meaning: the authoritative event for which the underlying interaction was merely production work. Meeting participants begin to attend on behalf of the minutes. Speech adjusts in advance to what can survive compression, and the durable record outcompetes the ambiguous, partially remembered encounter from which it was made. The summary no longer compresses the meeting; the meeting expands the summary.

## Maintenance

Inherited meaning: the work required to preserve an operational reality. Displaced meaning: the category into which unmediated access, accountability, and other expensive defaults are moved before they disappear from ordinary use. A real window becomes a maintenance liability beside a brighter display. A named remedy becomes operational overhead beside a frictionless apology. Nothing has been prohibited; it has merely become too costly, exceptional, or inconvenient to select.

## 10. Capability, Contamination, and the Human Baseline

Rhetorical debugging must extend from individual terms to measurement architecture. A benchmark score is not self-interpreting. It depends on the relation between training and test data, the number of optimization attempts, the resources allocated to each side, the conditions under which human baselines were collected, and the distance between the proxy task and the capacity named in the press release.

Three separations are therefore required. First, distinguish task performance from the mechanism producing it: retrieval, memorization, tool use, and generalization can yield the same answer while supporting different extrapolations. Second, distinguish the best system run from the distribution of deployments: selecting a successful sample and comparing it with an average human reverses ordinary experimental caution. Third, distinguish benchmark closure from capability closure. Saturating a finite question set demonstrates that the optimization process has captured the test; it does not by itself show that the open-ended phenomenon named by the test has been exhausted.

The appropriate correction is not to dismiss leaderboards, but to restore provenance. A serious report states exposure controls, inference budgets, selection procedures, human compensation and timing, uncertainty intervals, failure coverage, and the conditions under which the ranking reverses. Without these, “surpassing humanity” is less a scientific conclusion than a choice of denominator.

## 11. The User as an Extractive Interface

The vocabulary of service often conceals a reversal in who is serving whom. CAPTCHA makes the reversal unusually visible. The machine interrupts a human and assigns a classification task so the human may prove membership in the category “human.” Whatever the contemporary data pipeline behind a particular CAPTCHA, the interaction retains the form of unpaid labeling labor: the boundary authority gives homework to the entity seeking passage. It is a toll booth that calls itself a conversation.

Error interfaces complete the asymmetry. The system may retain a precise internal exception, trace identifier, component name, and timestamp while displaying “Something went wrong.” Diagnostic specificity is withheld at exactly

the point where it would enable comparison, escalation, or the naming of an accountable subsystem. The user is then asked for rich feedback. Information flows inward at maximum resolution and outward at minimum resolution.

Even apparently benign completion states participate in this architecture. “All caught up” can function less as an endpoint than as a temporary rest screen before held-back content is introduced. The interface produces the sensation of completion without permitting departure from the extraction loop. The debugging question is not whether each design choice is malicious. It is whether the information and control gradients consistently point toward retention.

## 12. Limit Cases: When the Record Replaces the World

The logic becomes clearest in speculative limit cases because fiction can hold the variables still. Consider a duplicated person whose memories, obligations, grief, and trauma continue across two bodies while an employer’s database permits only one employee record. The organization may assign patterned liabilities to both persons while recognizing benefits for only one because its foreign-key constraint accepts a single identity. The thought experiment exposes a general rule: administrative ontology becomes metaphysics when only what the database can represent is allowed to have institutional consequences.

Consider, next, Earth designated as a protected historical site. Carbon charges, heritage rules, and conservation levies may each be individually intelligible, yet their composition could price ordinary inhabitants out of the world being protected. Former residents return as paid performers of authentic terrestrial life for orbital visitors. Protection succeeds at preserving the referent and fails at preserving participation in it. The price of safety is paid by the people whose presence made the object worth protecting.

Finally, imagine a station whose observation ports have been sealed behind panels because high-resolution displays provide a clearer view at lower maintenance cost. The rendered starfield is better by every specified measure and worse in one excluded dimension: photons from the represented stars no longer reach the viewer. Reality survives in the schematics as a premium or maintenance category. The example is not an argument against displays. It identifies the decision hidden inside the metric: the system first defines every value the substitute improves and then concludes that nothing was lost.

A related thought experiment concerns artificial hesitation. A shipboard computer begins to stutter under thermal throttling, and its users find the slowed responses more companionable because effort has become perceptible. When the system returns to instant operation, it seems less alive. The episode reverses the usual benchmark: inefficiency supplies the social evidence that correctness alone lacks. It warns that personification is not produced solely by corporate language. Users may recruit friction as evidence of interiority, preferring shared struggle to perfect service. A complete rhetoric of artificial systems must therefore analyze both manufactured attribution and the human appetite for it.

These limit cases converge on a single architectural danger. The record,

rendering, or administrative schema begins as a witness to reality, becomes the only admissible evidence of reality, and finally reorganizes reality so that the witness is never contradicted. Linguistic displacement is the early stage of that process. A word is first detached from its costly mechanism; institutions then optimize for the detached word; eventually the mechanism appears anomalous because it no longer fits the system built in its name.

This protocol does not presume that every optimistic claim is false, every safety team is insincere, or every metaphor is manipulative. Its purpose is narrower and more exact: to determine whether a narrative remains invariant when its consequences become inconvenient. The result is an audit of argumentative procedure rather than an automatic verdict on the underlying technology.

The deepest defect in the vocabulary of acceleration is thus not exaggeration but selective measurement. These arguments treat the imagined future as a fixed moral debt while treating the tangible present as a disposable draft. They preserve words after evacuating their requirements, invoke minds to claim achievement and weather to distribute blame, and install humans where responsibility must be recorded rather than where judgment can be exercised.

True technological advancement is compatible with precision because it does not require the ruler to change length between measurements. When a narrative depends on semantic inertia—keeping words in circulation after their referents have departed—it is not merely describing progress. It is defending a valuation. Rhetorical debugging reclaims the ability to measure the world as it is rather than as the ledger requires it to be.

## Companion Essays

This essay consolidates a series of twenty shorter pieces, each treating one mechanism, one phrase, or one thought experiment at length. They are listed here by register rather than by argument, for a reader who wants the single-case version of anything compressed above.

### Mechanism Essays

- Compute as Deterrent – the verification gap between nuclear material and computation.
- AGI by [Year] – the moving target that lets a forecast survive every miss.
- Safety Reports to Revenue – oversight without an external source of authority.
- Benchmark Leaderboards – the collapse of exam-writer, test-taker, and grader into one party.
- We Take This Seriously – a commitment's grammar without its components.
- The Emergency That Requires a Product Launch – self-authorization dressed as urgency.

- Open, Except for the Open Parts – the stepwise dilution of a load-bearing word.
- The Model Did It – personification granted on success and withdrawn on harm.
- The Cost of Not Building It – an accounting ledger with two different evidentiary standards.
- The Human in the Loop – responsibility assigned to a station rather than a judgment.

## **Interface Essays**

- Please Verify That You're Human – suspicion that runs in only one direction.
- Your Meeting Has Been Summarized – a record that outcompetes the event it was drawn from.
- Something Went Wrong – diagnostic precision withheld at the point it would be actionable.
- You're ALL Caught Up – completion rendered as a loading state.
- We Value Your Feedback – dissatisfaction collected as data rather than treated as a defect.

## **Limit Cases**

- The Ship's Computer – friction mistaken for interiority.
- There Are No Windows on the Space Station – reality demoted to a maintenance category.
- The Clone Has No Legal Standing – metaphysics settled by a database schema.
- Earth Is a Protected Historical Site – a rescue that prices out the people it rescues.
- The Last Human Job – a button pressed by someone with no way to know what it does.

## Appendices

### A. A Formal Model of Rhetorical Asymmetry

The diagnostic argument in the main text can be stated more generally as a failure of invariance under reversal of institutional valence. Let a claim be represented by

$$c = (p, e, a, t, m),$$

where  $p$  is its proposition,  $e$  its evidentiary support,  $a$  the actor to whom agency is attributed,  $t$  its temporal horizon, and  $m$  its estimated magnitude. Let

$$s(c) \in \{-1, +1\}$$

denote the institutional valence of accepting the claim, with  $+1$  representing a claim favorable to deployment and  $-1$  a claim unfavorable to deployment.

Define an admissibility functional

$$\Phi(c) = w_e E(e) + w_m M(m) - w_t T(t) - w_u U(c),$$

where  $E$  measures evidentiary strength,  $M$  the magnitude assigned to the claimed consequence,  $T$  temporal distance, and  $U$  residual uncertainty. Under a procedurally symmetric evaluation regime, a claim is admitted when

$$\Phi(c) \geq \theta,$$

for a threshold  $\theta$  that is independent of  $s(c)$ .

Acceleration rhetoric instead permits

$$\theta = \theta(s),$$

with

$$\theta(-1) > \theta(+1).$$

Deployment-supporting claims therefore require less evidentiary support than otherwise comparable deployment-inhibiting claims. Define the threshold asymmetry

$$\mathcal{A}_\theta = \theta(-1) - \theta(+1).$$

A symmetric procedure has

$$\mathcal{A}_\theta = 0.$$

Positive values indicate an evidentiary advantage for deployment-supporting claims.

This formulation also permits asymmetry to enter through the weights rather than the threshold. Let

$$\mathbf{w}_s = (w_{e,s}, w_{m,s}, w_{t,s}, w_{u,s})$$

be the weighting vector applied under valence  $s$ . Then a stronger invariance criterion is

$$\mathbf{w}_+ = \mathbf{w}_- \quad \text{and} \quad \theta(+1) = \theta(-1).$$

The important point is that rhetorical asymmetry need not consist of a speaker making a false claim. It may instead consist of evaluating two true or uncertain claims with different rulers.

## A.1 The Valence-Reversal Operator

Define a reversal operator  $R$  that preserves the epistemic structure of a claim while reversing its institutional consequence:

$$R : (p, e, a, t, m, +1) \mapsto (p', e', a', t', m', -1),$$

subject to the matching conditions

$$E(e') = E(e), \quad T(t') = T(t), \quad M(m') = M(m), \quad U(Rc) = U(c).$$

A procedurally invariant evaluator  $D$  should satisfy

$$D(c) = D(Rc),$$

where  $D(c) \in \{0, 1\}$  denotes rejection or admission after controlling for the sign of the consequence.

This yields a direct empirical criterion. For a corpus containing  $N$  matched claim pairs, define

$$\widehat{\mathcal{I}} = \frac{1}{N} \sum_{i=1}^N |D(c_i) - D(Rc_i)|.$$

Here  $\widehat{\mathcal{I}} = 0$  indicates perfect procedural invariance, whereas  $\widehat{\mathcal{I}} = 1$  indicates that every matched pair changes admissibility when its institutional valence is reversed.

The statistic does not determine which conclusion is correct. It determines whether the procedure for reaching conclusions survives a change in which conclusion is convenient.

## B. Agency Conservation and the Personification Pivot

The personification pivot can be formalized as a change in causal resolution between favorable and unfavorable outcomes.

Let the causal description of outcome  $o$  be a probability distribution

$$P(A = a \mid o)$$

over candidate agents  $a \in \mathcal{G}$ , where  $\mathcal{G}$  may include the model, developer, deploying organization, operator, user, environment, and other relevant actors. Define the attribution vector

$$\mathbf{p}(o) = (P(A = a_1 \mid o), \dots, P(A = a_k \mid o)).$$

For matched successful and harmful outcomes  $o^+$  and  $o^-$ , define the agency displacement

$$\Delta_A = d(\mathbf{p}(o^+), \mathbf{p}(o^-)),$$

where  $d$  may be total variation distance,

$$d_{\text{TV}}(P, Q) = \frac{1}{2} \sum_{a \in \mathcal{G}} |P(a) - Q(a)|.$$

A large  $\Delta_A$  is not by itself evidence of rhetorical manipulation: different outcomes can genuinely have different causes. The diagnostic condition is stronger. Let  $X$  denote the underlying causal facts known about the system. Then define

$$\Delta_A^* = d_{\text{TV}}(P(A \mid o^+, X), P(A \mid o^-, X)).$$

If the causal architecture  $X$  remains substantially unchanged while attribution migrates systematically with outcome valence, the narrative exhibits an agency pivot.

A simpler corpus-level statistic can be constructed from grammatical subjects. Let

$$H(o) = \begin{cases} 1, & \text{if a model is represented as an intentional agent,} \\ 0, & \text{otherwise,} \end{cases}$$

and let

$$O(o) = \begin{cases} 1, & \text{if a responsible organization or decision-maker is named,} \\ 0, & \text{otherwise.} \end{cases}$$

Then define

$$\Pi = [P(H = 1 \mid o^+) - P(H = 1 \mid o^-)] + [P(O = 1 \mid o^-) - P(O = 1 \mid o^+)].$$

A strongly positive  $\Pi$  describes precisely the rhetorical pattern at issue: intentional personification concentrates around success while organizational attribution concentrates around failure, allowing achievement to appear autonomous while liability remains diffuse.

The null hypothesis is not that identical nouns must appear in every sentence.

It is that causal granularity should not systematically covary with whether an outcome is commercially useful.

## C. Forecasts, Calibration, and the Accelerating Miss

A dated technological forecast should ordinarily expose its issuer to epistemic loss. Let a forecast issued at time  $t_0$  assign probability

$$p = P(E \leq T \mid \mathcal{F}_{t_0})$$

to event  $E$  occurring before deadline  $T$ . Once  $T$  passes, let

$$y = \begin{cases} 1, & E \text{ occurred by } T, \\ 0, & E \text{ did not occur by } T. \end{cases}$$

A proper scoring rule, such as the Brier score,

$$B(p, y) = (p - y)^2,$$

penalizes confident misses. For  $y = 0$ ,

$$B(p, 0) = p^2,$$

so a failed 0.9-confidence forecast incurs a larger loss than a failed 0.55-confidence forecast.

The “accelerating miss” reverses this relation rhetorically. Let  $U_t$  represent the urgency attributed to development at time  $t$ . Under ordinary forecast accountability one expects, *ceteris paribus*,

$$\frac{\partial U}{\partial B} \leq 0 :$$

Larger forecasting error should not itself increase confidence in the forecasting framework.

The accelerating-miss mechanism instead behaves as

$$\frac{\partial U}{\partial B} > 0.$$

The forecast fails, but the failure is redescribed as evidence that the field is changing too rapidly for ordinary prediction, thereby increasing the urgency that the forecast originally supported.

For a sequence of forecasts  $(p_i, T_i, y_i)$ , define cumulative forecasting loss

$$L_n = \sum_{i=1}^n (p_i - y_i)^2$$

and cumulative rhetorical urgency  $U_n$ . A particularly strong empirical signature of the accelerating miss would be

$$\text{Cov}(\Delta L_n, \Delta U_n) > 0$$

over repeated misses: prediction error and asserted urgency rise together.

This criterion distinguishes revision from ratcheting. A legitimate failed forecast may produce a new model, new evidence, and a revised probability. The pathological case is one in which error itself becomes evidence for the same directional conclusion.

## D. Human Oversight as a Queueing Problem

The phrase “human in the loop” can be evaluated independently of whether a human is physically present. Suppose decisions arrive for review according to a process with mean rate

$$\lambda$$

and each meaningful review requires mean service time

$$\tau.$$

A reviewer therefore has service rate

$$\mu = \frac{1}{\tau}.$$

For  $c$  independent reviewers, nominal utilization is

$$\rho = \frac{\lambda}{c\mu} = \frac{\lambda\tau}{c}.$$

A necessary stability condition is

$$\rho < 1.$$

If

$$\rho \geq 1,$$

the review queue cannot remain bounded in expectation. Meaningful review must therefore be replaced by delay, dropping, sampling, or cursory approval.

Even  $\rho < 1$  is insufficient for effective oversight. Let  $I \in [0,1]$  measure informational adequacy,  $K \in [0,1]$  reviewer competence,  $P \in [0,1]$  institutional protection for refusal, and  $C \in [0,1]$  causal control—the probability that a refusal actually changes the downstream action. Define effective oversight capacity

$$\Omega = (1 - \rho)_+ IKPC,$$

where

$$(x)_+ = \max(x, 0).$$

This multiplicative form is intentionally severe. If any indispensable condition approaches zero, effective oversight approaches zero even when the other conditions are excellent.

A human may therefore be present while

$$\Omega \approx 0.$$

This distinguishes human presence from human authority.

## D.1 Refusal Bandwidth

Let  $R$  denote the set of recommendations presented to human reviewers and let  $r(x) \in \{0, 1\}$  indicate refusal. Define observed refusal frequency

$$f_R = \frac{1}{|R|} \sum_{x \in R} r(x).$$

A low  $f_R$  does not by itself establish theatrical oversight; a good system may rarely require refusal. The stronger test intervenes on cases near the decision boundary. Let  $R_\epsilon$  denote deliberately constructed cases within distance  $\epsilon$  of the organization's stated prohibition threshold. Then the relevant quantity is

$$f_R(\epsilon) = P(r(x) = 1 \mid x \in R_\epsilon).$$

If reviewers possess nominal authority but reliably approve even controlled cases satisfying the organization's own refusal conditions, the discrepancy between formal and causal authority becomes experimentally observable.

## E. The Counterfactual Brake

An oversight institution can be modeled as a control system. Let  $x_t$  represent the state of a deployment process at time  $t$ , and let

$$u_t \in \{\text{continue}, \text{modify}, \text{halt}\}$$

represent the action selected by the governing institution.

Let an oversight body emit finding  $z_t$ . Operational oversight requires that there exist at least one admissible finding  $z^*$  such that

$$P(u_t = \text{halt} \mid z_t = z^*) > P(u_t = \text{halt} \mid z_t = z_0),$$

where  $z_0$  denotes an ordinary or favorable finding.

The stronger counterfactual-brake condition requires

$$P(u_t = \text{halt} \mid z_t = z^*, E = \text{continue}) > 0,$$

where  $E = \text{continue}$  denotes executive preference for deployment.

Thus an oversight body possesses causal braking power only if some finding can alter the deployment state against the preference of the actor being overseen.

Define brake effectiveness

$$\mathcal{B} = \sup_z [P(u = \text{halt} \mid z, E = \text{continue}) - P(u = \text{halt} \mid z_0, E = \text{continue})].$$

If

$$\mathcal{B} = 0,$$

then no possible output of the oversight process changes the relevant decision. The body may still collect information, improve documentation, or advise management, but it is not a brake in the causal sense used here.

One may additionally define institutional durability  $D$  as the probability that the oversight body's authority survives an adverse finding:

$$D = P(\text{authority persists at } t+1 \mid z_t = z^*).$$

A body with large nominal  $\mathcal{B}$  but  $D \approx 0$  has a single-use brake: it can stop one release only by destroying its own ability to stop the next.

## F. Semantic Dilution as Requirement Loss

Prestige terms can be represented by sets of operational requirements. Let

$$\mathcal{R}_0(w) = \{r_1, r_2, \dots, r_n\}$$

denote the requirements associated with term  $w$  at a reference time. At later time  $t$ , let

$$\mathcal{R}_t(w) \subseteq \mathcal{R}_0(w)$$

denote the requirements still necessary for an artifact to receive the same label.

Define requirement retention

$$Q_t(w) = \frac{|\mathcal{R}_t(w)|}{|\mathcal{R}_0(w)|}.$$

Then semantic dilution is represented by

$$Q_t(w) < 1,$$

and progressive requirement shedding by

$$\frac{dQ_t(w)}{dt} < 0.$$

This alone is ordinary semantic change. The stronger phenomenon described in the essay occurs when reputational value does not fall proportionally with requirement retention.

Let  $V_t(w)$  denote the institutional or reputational value obtained by using the term. Define semantic leverage

$$\Lambda_t(w) = \frac{V_t(w)}{Q_t(w) + \varepsilon},$$

where  $\varepsilon > 0$  prevents division by zero.

If requirements disappear while prestige remains approximately constant,

$$Q_t(w) \downarrow, \quad V_t(w) \approx V_0(w),$$

then

$$\Lambda_t(w) \uparrow.$$

The term is doing increasing institutional work per surviving operational requirement.

For “open,” for example, one might define a requirement vector

$$r = (r_{\text{source}}, r_{\text{weights}}, r_{\text{data}}, r_{\text{reproduction}}, r_{\text{modification}}, r_{\text{redistribution}}).$$

The purpose of the vector is not to decree a single correct definition of openness. It makes definitional change auditable. A speaker may propose a different vector or weighting, but the requirements that disappeared can no longer disappear silently.

## G. The Rigged Ledger as Discounting Asymmetry

The treatment of future benefits and present harms can be expressed as an asymmetric valuation problem.

Let a consequence  $i$  have estimated magnitude  $x_i$ , probability  $p_i$ , and temporal delay  $t_i$ . Under a common exponential discount rate  $\delta$ , its expected present value is

$$V_i = p_i x_i e^{-\delta t_i}.$$

A procedurally symmetric ledger uses the same probability discipline and discounting rule regardless of whether  $x_i$  is positive or negative.

Suppose instead that benefits and harms receive separate parameters:

$$V = \sum_{i \in B} \hat{p}_i^B x_i e^{-\delta_B t_i} - \sum_{j \in H} \hat{p}_j^H |x_j| e^{-\delta_H t_j}.$$

The rhetoric described in the main text corresponds to the tendency

$$\hat{p}_i^B > p_i, \quad \hat{p}_j^H < p_j,$$

together with an unusual temporal treatment in which distant benefits are discounted weakly while immediate harms are rhetorically discounted as temporary:

$$\delta_B < \delta_H^{\text{rhet}},$$

where  $\delta_H^{\text{rhet}}$  denotes not literal financial discounting but the rate at which documented harms are treated as transient “implementation details.”

Define ledger asymmetry

$$\mathcal{L} = \left( \overline{\hat{p}^B} - p^B \right) - \left( \overline{\hat{p}^H} - p^H \right).$$

Large positive  $\mathcal{L}$  indicates simultaneous probability inflation for benefits and probability suppression for harms.

The important methodological consequence is that disagreement over the final sum  $V$  should come after inspection of the accounting rules. If the two columns use different probability estimators, different temporal rules, and different standards of evidence, their arithmetic difference has no stable interpretation.

## H. A Coding Protocol for Empirical Testing

The preceding quantities can be estimated from a corpus of press releases, technical reports, policy testimony, incident reports, interviews, and product documentation.

Divide the corpus into individual claims  $c_i$ . For each claim, record a coding vector

$$\mathbf{x}_i = (s_i, e_i, t_i, a_i, g_i, r_i, b_i),$$

where  $s_i$  is institutional valence,  $e_i$  evidentiary strength,  $t_i$  temporal horizon,  $a_i$  attributed agent,  $g_i$  causal granularity,  $r_i$  requirements attached to prestige terminology, and  $b_i$  the presence or absence of a genuine counterfactual brake.

The central hypotheses of the essay can then be stated as testable inequalities:

$$E[e_i \mid s_i = -1] > E[e_i \mid s_i = +1]$$

for the evidentiary threshold;

$$P(\text{intentional model language} \mid s_i = +1) > P(\text{intentional model language} \mid s_i = -1)$$

for asymmetric personification;

$$P(\text{named organizational responsibility} \mid s_i = -1) < P(\text{named organizational responsibility} \mid s_i = +1)$$

when failure language evacuates institutional agency; and

$$P(b_i = 1 \mid \text{“safety” language}) < 1$$

to distinguish invocation of safety from possession of causal stopping authority.

Because many of these variables require interpretation, inter-rater agreement should be reported. For two coders and categorical variable  $X$ , Cohen’s kappa

is

$$\kappa = \frac{p_o - p_e}{1 - p_e},$$

where  $p_o$  is observed agreement and  $p_e$  agreement expected by chance.

The empirical project therefore does not require proving that a particular speaker intended to manipulate language. The relevant object is the distributional pattern. If independent coders repeatedly find that standards of evidence, grammatical agency, semantic requirements, and oversight authority move in the predicted directions with institutional valence, the claim becomes one about an observable rhetorical system rather than inferred motive.

## I. Failure Conditions and Falsification

The framework should itself be subjected to the symmetry it demands from its objects. Several observations would weaken or falsify its strongest claims.

First, if matched deployment-supporting and deployment-inhibiting claims are admitted under statistically indistinguishable evidentiary thresholds, then

$$\mathcal{A}_\theta \approx 0,$$

and the evidentiary-asymmetry hypothesis is unsupported.

Second, if attribution remains stable across success and failure after controlling for genuine causal differences, then

$$\Delta_A^* \approx 0$$

and the personification-pivot hypothesis fails.

Third, if internal safety bodies routinely retain protected authority after blocking commercially important releases, then both

$$B > 0 \quad \text{and} \quad D \approx 1,$$

contradicting the strongest version of the institutional-theater thesis.

Fourth, if prestige attached to terms such as “open” declines approximately in proportion to the operational requirements removed from them, then semantic leverage remains stable:

$$\frac{d\Lambda_t}{dt} \approx 0.$$

In that case the phenomenon is better described as ordinary semantic change than reputation borrowing.

Finally, if failed forecasts reliably reduce subsequent confidence and urgency unless new independent evidence appears, then

$$\text{Cov}(\Delta L, \Delta U) \leq 0,$$

and the accelerating-miss mechanism is absent.

These conditions matter because rhetorical debugging cannot coherently demand falsifiability from technological narratives while insulating its own diagnoses from counterevidence. Its strongest claim is therefore conditional: where the proposed invariances fail systematically and in the predicted direction, linguistic displacement is doing measurable institutional work. Where they do not fail, the vocabulary may be vivid, imprecise, fashionable, or annoying without constituting the mechanism described here.

## References

- [1] Solon Barocas and Andrew D. Selbst, “Big Data’s Disparate Impact,” *California Law Review*, vol. 104, no. 3, pp. 671–732, 2016. doi: 10.15779/Z38BG31.
- [2] Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell, “On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?” in *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT ’21)*, pp. 610–623, 2021. doi: 10.1145/3442188.3445922.
- [3] Rishi Bommasani et al., “On the Opportunities and Risks of Foundation Models,” *arXiv preprint arXiv:2108.07258*, 2021. doi: 10.48550/arXiv.2108.07258.
- [4] Joy Buolamwini and Timnit Gebru, “Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification,” in *Proceedings of the 1st Conference on Fairness, Accountability and Transparency, Proceedings of Machine Learning Research*, vol. 81, pp. 77–91, 2018.
- [5] Glenn W. Brier, “Verification of Forecasts Expressed in Terms of Probability,” *Monthly Weather Review*, vol. 78, no. 1, pp. 1–3, 1950. doi: 10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2.
- [6] A. P. Dawid, “The Well-Calibrated Bayesian,” *Journal of the American Statistical Association*, vol. 77, no. 379, pp. 605–610, 1982. doi: 10.1080/01621459.1982.10477856.
- [7] Finale Doshi-Velez et al., “Accountability of AI Under the Law: The Role of Explanation,” *arXiv preprint arXiv:1711.01134*, 2017. doi: 10.48550/arXiv.1711.01134.
- [8] Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daume III, and Kate Crawford, “Datasheets for Datasets,” *Communications of the ACM*, vol. 64, no. 12, pp. 86–92, 2021. doi: 10.1145/3458723.
- [9] Charles Goodhart, “Problems of Monetary Management: The U.K. Experience,” in A. S. Courakis (ed.), *Inflation, Depression, and Economic Policy in the West*, London: Mansell, pp. 111–146, 1981.
- [10] Zachary C. Lipton, “The Mythos of Model Interpretability,” *Queue*, vol. 16, no. 3, pp. 31–57, 2018. doi: 10.1145/3236386.3241340.
- [11] Margaret Mitchell, Simone Wu, Andrew Zaldivar, Parker Barnes, Lucy Vasserman, Ben Hutchinson, Elena Spitzer, Inioluwa Deborah Raji, and Timnit Gebru, “Model Cards for Model Reporting,” in *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT\* ’19)*, pp. 220–229, 2019. doi: 10.1145/3287560.3287596.

- [12] Clifford Nass, Jonathan Steuer, and Ellen R. Tauber, "Computers Are Social Actors," in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, pp. 72-78, 1994. doi: 10.1145/191666.191703.
- [13] Ivo Adan and Jacques Resing, *Queueing Theory*. Eindhoven University of Technology, 2002.
- [14] Raja Parasuraman and Dietrich H. Manzey, "Complacency and Bias in Human Use of Automation: An Attentional Integration," *Human Factors*, vol. 52, no. 3, pp. 381-410, 2010. doi: 10.1177/0018720810376055.
- [15] Raja Parasuraman, Thomas B. Sheridan, and Christopher D. Wickens, "A Model for Types and Levels of Human Interaction with Automation," *IEEE Transactions on Systems, Man, and Cybernetics-Part A: Systems and Humans*, vol. 30, no. 3, pp. 286-297, 2000. doi: 10.1109/3468.844354.
- [16] Inioluwa Deborah Raji, Andrew Smart, Rebecca N. White, Margaret Mitchell, Timnit Gebru, Ben Hutchinson, Jamila Smith-Loud, Daniel Theron, and Parker Barnes, "Closing the AI Accountability Gap: Defining an End-to-End Framework for Internal Algorithmic Auditing," in *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*, pp. 33-44, 2020. doi: 10.1145/3351095.3372873.
- [17] Andrew D. Selbst, Danah Boyd, Sorelle A. Friedler, Suresh Venkatasubramanian, and Janet Vertesi, "Fairness and Abstraction in Sociotechnical Systems," in *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT\* '19)*, pp. 59-68, 2019. doi: 10.1145/3287560.3287598.
- [18] Ben Shneiderman, "Human-Centered Artificial Intelligence: Reliable, Safe & Trustworthy," *International Journal of Human-Computer Interaction*, vol. 36, no. 6, pp. 495-504, 2020. doi: 10.1080/10447318.2020.1741118.
- [19] Harini Suresh and John V. Guttag, "A Framework for Understanding Sources of Harm throughout the Machine Learning Life Cycle," in *Proceedings of the 1st ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization*, 2021. doi: 10.1145/3465416.3483305.
- [20] Joseph Weizenbaum, "ELIZA-A Computer Program for the Study of Natural Language Communication Between Man and Machine," *Communications of the ACM*, vol. 9, no. 1, pp. 36-45, 1966. doi: 10.1145/365153.365168.
- [21] Joseph Weizenbaum, *Computer Power and Human Reason: From Judgment to Calculation*. San Francisco: W. H. Freeman, 1976.
- [22] Christopher D. Wickens, John D. Lee, Yili Liu, and Sallie E. Gordon-Becker, *An Introduction to Human Factors Engineering*, 2nd ed. Upper Saddle River, NJ: Pearson, 2004.
- [23] George Lakoff and Mark Johnson, *Metaphors We Live By*. Chicago: University of Chicago Press, 1980.

- [24] Open Source Initiative, “The Open Source Definition,” version 1.9. Available from the Open Source Initiative.
- [25] Open Source Initiative, “The Open Source AI Definition,” version 1.0, 2024.
- [26] National Institute of Standards and Technology, Artificial Intelligence Risk Management Framework (AI RMF 1.0). NIST AI 100-1, 2023. doi: 10.6028/NIST.AI.100-1.
- [27] European Union, “Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act),” Official Journal of the European Union, 2024.